The Dollar's Safe-Haven Status Across Two Shocks: Evidence from Liberation Day and the Strait of Hormuz Crisis

Bachelor's Thesis

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1 Introduction

Within a single year, two major shocks moved global commodity and financial markets. On the 2nd of April 2025, widely referred to as ”Liberation Day”, the United States Government announced broad-based tariffs on virtually all of its trading partners, citing persistent trade deficits and national security concerns. The policy included a base 10% tariff and, subsequently, additional higher rates for trading partners with large surpluses against the U.S., such as China and the European Union. Following the announcement, the dollar fell sharply against a broad basket of currencies, with the Euro, Yen and Swiss Franc appreciating against it—the Franc by more than 3% within the first week. The move was accompanied by a spike in equity-market volatility and a U.S. equities sell-off.

Less than a year later, on 28 February 2026, the United States and Israel launched airstrikes on Iran, beginning the 2026 Iran war. As escalation in the Gulf threatened to close the Strait of Hormuz and sent crude sharply higher, the dollar did exactly what the safe-haven literature predicts—it appreciated.

The puzzle speaks to a live debate about whether the dollar's exceptional status is eroding. A growing literature argues that the currency's safe-haven premium has weakened, pointing to the unusual co-movement of a falling dollar, rising Treasury yields and rising risk premia during the 2025 tariff episode (Ostry et al. 2025; Kamin 2025; Jiang et al. 2026). Against this, the long tradition on the dollar's "exorbitant privilege" and its role as the world's reserve and safe-haven currency (Gourinchas and Rey 2005; Obstfeld and Zhou 2023; Habib and Stracca 2012; Ranaldo and Söderlind 2010) implies the status should be durable. The two 2025–2026 shocks offer an unusually clean natural experiment for adjudicating the debate, because they differ in *kind*: one is a U.S. trade-policy shock, the other a geopolitical oil-supply shock.

The central question of this thesis is therefore: *does the dollar's safe-haven response depend on the type of shock—even when the United States is involved in both?* To adjudicate it, the design sets the two shocks against a historical benchmark—three episodes in all. The Liberation Day tariff announcement is a purely U.S.-originated economic-policy shock. The 2026 Israel–Iran/Hormuz crisis is a geopolitical oil-supply shock in which the United States is involved as a co-belligerent but still functions as a financial safe harbour. And the 2022 Russian invasion of Ukraine—an external military/oil shock not authored by the United States—serves as a control for the "classical" safe-haven response. Comparing the three isolates whether the dollar's behaviour is conditioned by the nature of the shock or merely by who caused it.

The thesis answers the question with two complementary empirical layers on a single daily dataset spanning January 2019 to June 2026. An *event study* measures the cumulative abnormal response of the dollar, a cross-section of currency baskets, gold, equities and crude oil around each shock, and tests formally whether the dollar's response differs across events. A complementary *regression* then tests whether the dollar's relationship with risk itself changed across the episodes—whether its safe-haven sensitivity to volatility held or broke. The two layers are complementary: the event study isolates the dollar's move around each discrete, dated shock, while the regression tests whether its underlying sensitivity to risk shifted across the episodes.

The contribution is threefold. First, to the best of the author's knowledge this is the first systematic side-by-side comparison of the Liberation Day and Iran/Hormuz episodes for currency markets. Second, the thesis bridges the safe-haven and commodity-currency literatures by carrying the oil channel and a gold benchmark through the same design, so that the dollar's behaviour can be read against gold and the oil-exposed currency cross-section rather than in isolation. Third, the rebuilt carry/dollar risk factors and the WMR currency panel extend standard cross-sectional tools through to mid-2026, allowing the most recent shocks to be studied with consistent, citable data.

The remainder of the thesis is organised as follows. Chapter 2 reviews the safe-haven, dollar-erosion and commodity-currency literatures. Chapter 3 sets out the conceptual framework—the working definition of a safe-haven currency, a classification that separates a shock's origin from its nature, and the hypotheses that follow. Chapter 4 narrates the 2022–2026 events. Chapter 5 describes the data and the event-study and regression methodologies. Chapter 6 presents the empirical results. Chapter 7 discusses the findings and their implications, and Chapter 8 concludes.

The results point to a dollar whose safe-haven status is conditional on the type of shock rather than on its originator. The two U.S.-involved shocks produce a mirror image: the Liberation Day tariffs depreciate the broad dollar index by over three and a half percent within twenty trading days, while the Hormuz oil-supply crisis appreciates it—by as much as the external Ukraine benchmark, from which its response is statistically indistinguishable. The complementary regression confirms the pattern: the dollar's risk sensitivity, significantly positive on normal days, collapses to essentially zero inside the tariff window yet holds through the Hormuz and Ukraine episodes. Gold emerges as a hedge against the dollar rather than a universal safe haven, and the oil channel sorts the currency cross-section only for the supply-driven shock.

2 Literature Review

This chapter situates the thesis within six distinct strands of literature. Each body of research addresses a specific dimension of the core question the thesis tries to answer: does the dollar's status as a safe haven depend on the nature of a shock, rather than its source? By reviewing these fields, this chapter establishes the theoretical framework to understand how the dollar functions as a safe haven, setting the stage for examining the consequences of when the United States itself originates a shock, acts as a co-belligerent or is not directly involved.

2.1 Currency Risk Factors

Lustig et al. (2011) establish the foundational framework for understanding currency risk by identifying two primary common risk factors, a "level" factor, the average excess return on foreign-currency portfolios against the dollar, and a "slope" factor, which captures the excess returns of carry trades. They demonstrate that high-interest-rate currencies load more heavily on the slope factor, which is closely linked to global equity-market volatility. Crucially, their model shows that during periods of high global risk, low-interest-rate currencies endogenously appreciate, acting as a hedge, while high-interest-rate currencies depreciate.

The dollar's special role is not confined to asset markets. It also dominates the invoicing of international trade. Drawing on evidence from more than 2,500 country pairs covering some ninety per cent of world trade, Gopinath et al. (2020) show that traded-goods prices are predominantly sticky in dollars rather than in the exporter's or the importer's currency. Trade prices and volumes therefore respond to the dollar exchange rate more than to the bilateral rate.

Building on this dominant-currency paradigm, Dalgic and Ozhan (2025) show that a currency's risk premium is not exogenous but rooted in an economy's trade and financial structure, namely the share of its trade invoiced in dollars and the amount of dollar debt it carries. In their small open-economy model, a shock that depreciates the local currency, such as a foreign interest-rate hike, turns contractionary rather than expansionary. Dollar debt opens a financial channel through which the depreciation raises the real value of liabilities. Servicing dollar debt now requires more local currency. Dollar invoicing mutes the trade channel, leaving export volumes largely unresponsive to the exchange rate, providing no offset. Because depreciations are therefore recessionary, the exchange rate turns countercyclical. The currency loses value precisely when domestic income falls, offers its holder no insurance, and is therefore a poor hedge against domestic downturns. Investors hold such currencies only against a sufficient risk premium, the mirror image of the safety commanded by the dollar. The dollar and carry factors, extended to June 2026, form the basis of the portfolio construction in Chapter 5.

2.2 Dollar Dominance and Its Erosion

The dollar's role as the primary global reserve currency confers an "exorbitant privilege", a persistent excess return on U.S. external assets over its liabilities (Gourinchas and Rey 2005). Gourinchas et al. (2017) show that this privilege is mirrored by an "exorbitant duty". In global crises the United States transfers wealth abroad, roughly 19% of U.S. GDP during 2007–2009. Because it holds risky foreign assets while owing safe dollar liabilities, collapsing global asset prices and the appreciating dollar both push its net foreign asset position downward, with the exchange rate accounting for roughly a third of the valuation effect. Jiang et al. (2021) find the source of that movement, showing that surges in foreign demand for safe dollar assets appreciate the dollar immediately. They capture the demand through the convenience yield on U.S. Treasuries, the return investors give up in exchange for safety and liquidity. The dollar's appreciation in a global crisis is therefore a consequence of its role as the world's supplier of safe assets. Obstfeld and Zhou (2023) add the cyclical dimension, describing a "global dollar cycle" in which dollar appreciation tightens financial conditions worldwide. They find that dollar movements correlate most reliably with measures of investor risk appetite.

While the 2008 financial crisis and the COVID-19 pandemic elicited the standard appreciation, Jiang et al. (2026) argue that the dollar has been losing that standing gradually, documenting convenience yields on dollar safe assets eroding over the two years before the 2025 tariff announcements. A comparable trend is evident in central bank reserves. Arslanalp et al. (2022) document a two-decade decline in the dollar's share of global reserves, and attribute it to deliberate diversification by reserve managers rather than to valuation effects. Between 1999 and 2021 the dollar's share of global reserves declined from 71% to 59%. A quarter of what left the dollar moved into the Chinese renminbi, the remaining three quarters into currencies of smaller economies that had rarely served as reserve assets before.

2.3 Safe-Haven Currencies and Gold

A central paradox, identified by Habib and Stracca (2012), is the dollar's appreciation during the 2008 global financial crisis, a crisis that originated in the United States. Analysing 52 currencies over almost 25 years, they find that safe-haven status is best predicted not by the interest-rate spread that the carry-trade literature emphasises but by a country's net foreign asset position, which signals how exposed it is to external stress. A currency that has hedged well in the past also tends to continue doing so, and the currencies of large but less financially open economies hedge better in periods of distress. The interest-rate spread against the United States, by contrast, matters only during the 2008 crisis itself.

Cheema et al. (2022) compare a range of safe-haven assets across different crises, and find their safe-haven character changed considerably during the COVID-19 pandemic compared with the 2008 financial crisis. The Swiss franc remained an intermediate haven in most settings, while gold and silver, weak havens in 2008, turned risky. The dollar deteriorated as well, from intermediate protection in 2008 to weak protection during the pandemic.

Providing broader context, Bock and de Carvalho Filho (2015) demonstrate that the dollar, Japanese yen and Swiss franc consistently appreciate against other currencies during risk-off episodes. How much each currency gains depends on its interest rate before the episode, its net foreign asset position, whether it is overvalued, and how cheaply and freely it can be traded. Ranaldo and Söderlind (2010) show that during U.S. equity sell-offs the franc and yen appreciate against the dollar, with effects that appear within hours, persist for several days, and grow more than proportionally as volatility rises. The dollar therefore strengthens broadly in risk-off episodes while still losing ground to the franc and the yen.

Baur and Lucey (2010) are the first to formally define and test the distinction between a hedge and a safe haven. In their terminology, a hedge is uncorrelated or negatively correlated with equities on average, which does not guarantee protection during market crashes. A safe haven must deliver exactly that protection in the extreme. Applying the distinction to gold, they find that it hedges equities on average, but works as a safe haven only after severe market downturns, and even then only for a short time. This strand supplies the thesis's working definition of a safe haven and its benchmarks. Chapter 3 adopts the conditional, crisis-contingent notion of safety, and the yen, the franc and gold serve as the standards against which the dollar's response is measured.

2.4 Geopolitical Risk

Caldara and Iacoviello (2022) provide the foundational measurement of geopolitical risk, a news-based index that separates geopolitical "threats", events that are feared but have not yet occurred, from geopolitical "acts", their actual realisation. Shocks to the index depress investment, employment and stock prices, and the damage persists, with both the threat and the realisation components contributing. Expanding this to exchange rates, Yilmazkuday (2025) demonstrates that the dollar's geopolitical safe-haven status is not universal. A year after a shock to global geopolitical risk, the U.S. dollar has depreciated significantly, even as currencies such as the South African rand or Brazilian real have appreciated. The currencies of countries most deeply integrated into international supply chains depreciate most, a pattern driven primarily by geopolitical acts rather than threats. Complementing this, Liu and Zhang (2024) find that a zero-cost strategy that buys the currencies of high-geopolitical-risk countries and sells those of low-risk countries earns a significant excess return of 5.72% per year, and that this return predictability stems from country-specific and regional risk components rather than a single global factor. A currency's risk–return profile is therefore sensitive to the geographic origin of the shock.

Institutional evidence points in the same direction. The International Monetary Fund (2025) finds that although geopolitical risk events exert only a modest effect on asset prices on average, major events, military conflicts in particular, produce substantially larger and more persistent equity declines and raise governments' borrowing costs. The effects are strongest in emerging markets with little fiscal room and limited foreign reserves, and they spill over to other countries through trade and financial ties.

The thesis draws on this strand directly. The index places the Hormuz and Ukraine episodes in their longer-run context in Chapter 4, its threat and act components are tracked around each shock in the event study, and changes in the index serve as a measure of geopolitical risk in the safe-haven regression.

2.5 Tariff Shocks and Exchange Rates

The 2025 tariff episode has generated a fast-growing literature that is central to this thesis. Kamin (2025) estimates the dollar's daily relationship with the VIX and interest differentials and finds a significant break after the Liberation Day announcements of April 2025. Its sensitivity to the VIX turns from positive to negative, so a currency that once gained when volatility spiked now falls with it, and investors no longer treated the dollar as a refuge in stress. Jiang et al. (2026) document the same break across a wider set of correlations. The dollar depreciated while U.S. interest rates rose relative to euro rates, the VIX climbed and the convenience yield on Treasuries fell, a pattern they attribute to weakening foreign demand for the safe assets the United States supplies. Hassan et al. (2025) provide the theoretical counterpart in a general-equilibrium model where a currency's safety arises endogenously. In their account the dollar is safe because disturbances originating in the United States move traded-goods prices worldwide, and a trade war that cuts the country off from world markets shuts down those spillovers, so the dollar's habit of strengthening in bad times fades and U.S. interest rates move higher.

The depreciation was not what the earlier record suggested. Khalil and Strobel (2024) show that the tariff rounds of 2018–2019 came with a stronger dollar and trace that appreciation to U.S. trade policy itself. Ostry et al. (2025) qualify the pattern with a database of tariff announcements, threats and implementations spanning 2018–2020 and 2025, weighting each shock by the size of the tariff and its economic relevance. The direction of the response, they find, hinges on retaliation. The dollar appreciates when the United States imposes tariffs unilaterally but depreciates when trading partners strike back, which leads them to conclude that the fall after 2 April 2025 was no anomaly once retaliation is taken into account. Kalemli-Özcan et al. (2026) offer a complementary explanation centred on uncertainty. In their model, doubt about future tariff policy feeds a risk premium into the exchange rate, and greater tariff uncertainty raises precautionary saving and the compensation investors demand, weakening the currency at once even though the tariffs alone would strengthen it. Calibrated to the 2025 episode, the model matches how far and how fast the dollar fell.

Benguria and Saffie (2025) measure the immediate market response to the 2 April announcement itself, separating the anticipated part of each country's tariff from the surprise created by rounding the rates upward. Only the surprise moves prices, a one-percentage-point higher tariff lowering U.S. equity prices by 0.23%, and the dollar shows no appreciation, if anything a depreciation against floating currencies. This strand supplies the thesis's first treatment. The Liberation Day shock enters the event study as the trade-policy episode against which the military and oil-supply shocks are compared.

2.6 Commodity Currencies and Oil

Y.-C. Chen and Rogoff (2003) show that for commodity exporters such as Australia, Canada and New Zealand, the world price of their exports is a strong driver of the real exchange rate. These three serve as the natural test cases, developed economies with floating exchange rates whose exports remain dominated by commodities. The effect is clearest for the Australian and New Zealand dollars. Kilian (2009) decomposes oil-price changes into supply disruptions, global demand for industrial commodities, and precautionary demand reflecting concern about future supply, each with distinct effects on the oil price, output and inflation. A supply-driven price increase is therefore not equivalent to a demand-driven one.

H. Chen et al. (2016) show that the dollar reacts more strongly to demand-driven than to supply-driven oil shocks. The reaction is largest against the Canadian dollar and the Norwegian krone, because Canada and Norway trade oil heavily with the United States. Oil has also become more important for the dollar over time. Before the global financial crisis, oil shocks accounted for 10% to 20% of the movement in the dollar's bilateral rates. Afterwards the share rises to over 40%, and for the Canadian dollar to more than half.

Albulescu and Ajmi (2021) document predictive causality between oil prices and the dollar in both directions, strongest in crisis periods. Salisu et al. (2022) show that oil tail risk predicts the tail risk of the dollar's rates against the Canadian dollar and the British pound, while the dollar-yen pair moves inversely, consistent with the yen's haven role.

For the thesis, this literature provides the basis for the oil-exporter versus oil-importer portfolio split in Chapter 5, and the distinction between supply-driven and demand-driven oil movements (Kilian 2009) separates the Hormuz crisis from the tariff shock. The episodes themselves, and the sources that document them, are covered in Chapter 4.

3 Conceptual Framework

This chapter turns the literature of Chapter 2 into the frame the empirical chapters apply. It fixes the working definition of a safe-haven currency, classifies the three shocks by origin and nature, maps both onto measurable quantities, and states the hypotheses.

3.1 Safe Haven, Hedge, and the Dollar's Conditional Status

Following Baur and Lucey (2010), this thesis adopts a conditional definition. A *hedge* is an asset that is uncorrelated or negatively correlated with risky assets on average. A *safe haven* is an asset that is uncorrelated or negatively correlated with risky assets in times of market stress. Safe-haven status is therefore a crisis-contingent property rather than an average one. Applied to currencies, its empirical signature is appreciation when global risk rises, that is when equity prices fall and volatility increases (Ranaldo and Söderlind 2010; Bock and de Carvalho Filho 2015).

The dollar's claim to that status is structural. The "exorbitant privilege" literature casts the United States as the centre of the international monetary system and the issuer of the assets the world treats as safe (Gourinchas and Rey 2005), and in global crises the flight into those assets appreciates the dollar (Gourinchas et al. 2017; Obstfeld and Zhou 2023). Across a broad currency panel, however, Habib and Stracca (2012) tie safe-haven behaviour to fundamentals such as the net foreign asset position rather than to interest-rate spreads. The question this thesis poses follows. Is the dollar's status unconditional, or can the shock itself suspend it, in particular when it originates in U.S. policy?

The definition carries two observable implications. Around a dated stress event a safe haven should record abnormal appreciation, which the event study measures episode by episode. On ordinary days its return should rise with risk, a sensitivity the regression of Chapter 6 tests inside each crisis window.

3.2 Classifying Shocks by Origin and Nature

The thesis separates two attributes that empirical work often confounds, who *originates* a shock and what *kind* of shock it is. The originator may be the United States or an external actor, the nature an economic-policy shock, here trade policy, or a military/oil-supply shock. Crossing the two yields the four cells of Table 3.1. The three episodes occupy three of them, and the fourth, an external trade-policy shock, has no counterpart in the sample.

Table 3.1: The three episodes in the origin–nature classification

	Trade-policy shock	Military/oil-supply shock
U.S.-linked origin	Liberation Day tariffs (2 April 2025, U.S.-originated)	Israel–Iran/Hormuz crisis (28 February 2026, U.S.-involved)
External origin	*no episode in sample*	Invasion of Ukraine (24 February 2022)

Notes. Dates are the event days that fix t_0 of the event windows in Chapter 5.

This crossing is what allows origin and nature to be told apart. If the dollar's response were governed by *origin*, the two U.S.-linked shocks would move the dollar alike and differ from the external Ukraine benchmark. If instead it were governed by *nature*, the two military/oil-supply shocks would resemble each other regardless of U.S. involvement, and the trade-policy shock would stand apart. The two readings make opposite predictions about which episodes cluster together, and the empirical chapters adjudicate between them.

The design's limits follow from the same table. Each occupied cell holds a single episode and the two U.S.-linked shocks differ in the degree of involvement, as originator in one case and co-belligerent in the other, so the comparison identifies patterns across these three episodes rather than properties of shock classes.

The oil move classifies the nature of a shock in its own right. Kilian (2009) distinguishes physical supply disruptions, global demand for industrial commodities, and precautionary demand driven by concern about future supply, each with different macroeconomic effects. In that scheme the Hormuz episode is broadly supply-side, lifting crude through disrupted shipments and precautionary demand, whereas the tariff shock depresses crude through weaker expected demand. Through the commodity-currency channel the crude response should also sort currencies by oil-trade position (Y.-C. Chen and Rogoff 2003; H. Chen et al. 2016).

3.3 From Concepts to Measurable Quantities

The concepts are measured on a small set of objects, constructed in Chapter 5. The dollar's response is read from the broad dollar index and an equal-weighted basket of the sample currencies, so no single bilateral rate drives the result. Following the currency risk-factor literature, a forward-discount sort yields safe funding and risky carry baskets (Lustig et al. 2011), and a split by net oil-trade position carries the oil channel into the cross-section. Gold, a crisis haven for equities (Baur and Lucey 2010), serves as the non-sovereign benchmark and is also read in euro terms to strip mechanical dollar effects. The VIX proxies market stress and the index of Caldara and Iacoviello (2022) tracks geopolitical tension.

3.4 Hypotheses

Together these strands yield one organising prediction, that the dollar's safe-haven response depends on the nature of a shock rather than on U.S. involvement. Three ex-ante hypotheses make it testable.

H1 (conditional safe haven). Around the military/oil-supply shocks, Ukraine and Hormuz, the dollar appreciates, as the safe-haven literature implies. Around the U.S.-originated trade-policy shock it depreciates, as the erosion literature suggests. Responses cluster by nature, not by U.S. involvement.

H2 (gold as the non-sovereign benchmark). Gold, free of sovereign and counterparty risk, has no reason to discriminate between shocks by origin. It retains its haven bid even in the U.S.-originated episode, where the erosion logic predicts a weaker dollar. Any conditionality under H1 is then specific to the dollar.

H3 (oil channel in the cross-section). The exporter–importer spread follows the sign of the crude move. Oil exporters outperform importers when a shock lifts crude, as under Hormuz and Ukraine, and lose ground when it depresses crude, as on Liberation Day.

These predictions are not foregone, since Yilmazkuday (2025) finds the dollar depreciating in the year after global geopolitical-risk shocks. H1 fails if the tariff episode moves like the military pair or the military pair splits along involvement, H2 if gold weakens in the U.S.-originated episode, H3 if the spread ignores the crude response. Chapter 6 reports the tests, Chapter 7 the verdicts.

4 Background: The Events of 2022–2026

This chapter establishes the institutional setting of the episodes the empirical chapters analyse and fixes the event dates that define the windows used in Chapter 5. For each episode it documents what happened, how official institutions characterised the market backdrop, and how the shock is placed within the origin–nature classification of Section 3.2; each episode section closes with the response the safe-haven hypothesis of Section 3.4 predicts *ex ante*, so that Chapter 6 reads as a test of stated predictions rather than a search for patterns. The scholarly debate these episodes feed into is reviewed in Chapter 2 and is not repeated here.

4.1 Liberation Day and the U.S. Tariff Shock

On 2 April 2025, in an announcement branded "Liberation Day," the United States imposed broad "reciprocal" tariffs on nearly all of its trading partners, combining a flat baseline rate with higher country-specific surcharges (Kamin 2025). The measure represented an abrupt change in the U.S. trade-policy regime rather than a marginal adjustment, and official assessments subsequently documented the resulting trade fragmentation and the pass-through of higher tariffs to economic activity (European Central Bank 2025; European Central Bank 2026).

The market reaction was one of broad risk aversion: equities sold off, implied volatility rose, and crude oil fell as market participants priced weaker global demand (European Central Bank 2025). Around the announcements, the dollar depreciated against the euro, yen and Swiss franc even as risk metrics deteriorated, and long-dated Treasury yields rose while the currency weakened (Kamin 2025). The competing interpretations of this break—a repricing of the dollar's reserve status and an erosion of its safety premium—are reviewed in Section 2.5.

Under the classification of Section 3.2, Liberation Day is the study's clearest U.S.-authored economic-policy shock, and the accompanying fall in crude marks it as a demand-side disturbance. A partial ninety-day tariff pause on 9 April 2025 tempered the equity drawdown without restoring the dollar (Kamin 2025), and is treated here as a separately dated reversal. Under the safe-haven hypothesis of Section 3.4, a U.S.-authored shock that erodes the dollar's safety premium should weaken the currency on announcement and recover, at most, only partially when the policy is partially walked back.

4.2 The Israel–Iran Escalation and the Strait of Hormuz

The Gulf episode unfolded in stages, from threat to realisation. A first, twelve-day confrontation in June 2025 – Israeli and U.S. strikes on Iranian facilities, an Iranian response, and a rapid ceasefire – lifted crude only briefly and left major currencies little changed (Kilian et al. 2026). Escalation resumed in early 2026 and turned acute on 28 February 2026, when the United States and Israel launched airstrikes on Iran. Iran retaliated and moved to close the Strait of Hormuz – the maritime chokepoint through which roughly a fifth of globally traded oil passes – declaring it closed on 2 March and claiming complete control on 4 March 2026 (Fattouh and Economou 2026). Attacks on shipping followed and Brent crude rose sharply, trading above $100 per barrel at the height of the disruption (International Monetary Fund 2026), with Kilian et al. (2026) tracing the inflationary consequences through the oil channel.

Under the classification of Section 3.2, the crisis is a military/oil-supply shock in which the United States is involved as a co-belligerent, and the rise in crude identifies it as a supply-driven disturbance – the mirror image of the demand-side tariff shock. Several further dated events punctuate the episode. A two-week ceasefire announced on 7 April 2026 committed to reopening the Strait, but collapsed within days when the Islamabad talks failed and the United States imposed a naval blockade on 13 April (International Monetary Fund 2026). A round of U.S. strikes followed on 25 May 2026, and the war was wound down only by a separate sixty-day ceasefire announced on 11 June 2026 (signed 17 June), which reopened the Strait. The April truce furnishes the within-crisis reversal the analysis exploits: under the safe-haven hypothesis of Section 3.4, a military/oil-supply shock should draw a haven bid into the dollar as the threat builds and release it on credible de-escalation, so that any premium the closure put into the dollar should come back out as the Strait is set to reopen. The June ceasefire falls too close to the end of the sample for a full event window (Section 6.5).

4.3 The Ukraine Control

The Russian invasion of Ukraine on 24 February 2022 serves as the external benchmark: a military/oil-supply shock the United States did not author and in which it was not a belligerent. The invasion combined a sharp spike in geopolitical risk with a supply-driven jump in energy prices, both visible in Figures 4.2 and 4.3. Episodes of this kind are the empirical reference point for the classical safe-haven response reviewed in Section 2.3: in risk-off events the dollar, yen and Swiss franc tend to appreciate against other currencies.

Because the United States is here a bystander rather than a protagonist, the invasion isolates the *nature* of the shock from U.S. authorship. Under the hypothesis of Section 3.4 the dollar should display the classical haven response, making the 2022 invasion the benchmark against which the U.S.-authored tariff shock and the U.S.-involved Hormuz crisis are compared.

4.4 The Structure of the Comparison

Taken together, these three episodes populate the origin–nature classification of Section 3.2 in a way that gives the comparison its identifying power. The invasion of Ukraine on 24 February 2022 is the external benchmark, a military/oil-supply shock the United States did not author; the Liberation Day tariffs of 2 April 2025 are the U.S.-authored trade-policy shock; and the Hormuz crisis, which began with the U.S.–Israeli airstrikes of 28 February 2026 and the closure of the Strait of Hormuz on 2 March, is a military/oil-supply shock in which the United States is involved as a co-belligerent. Throughout the empirical chapters, day 0 of the Hormuz crisis is 28 February 2026, the onset of the airstrikes; the closure declaration of 2–4 March enters the analysis only as a sub-event. Holding the crisis character of the shock roughly fixed while varying its origin is what allows the dollar's response to be attributed to the *nature* of a shock rather than to U.S. authorship, or the reverse: if origin governs the response, the two U.S.-linked episodes should move the dollar alike and apart from Ukraine, whereas if nature governs it, the two military/oil-supply shocks should resemble each other regardless of who authored them.

Around each treatment sits a dated reversal that sharpens the reading. The ninety-day tariff pause of 9 April 2025 partially unwinds the Liberation Day shock, and the two-week ceasefire of 7 April 2026 unwinds the Hormuz threat; if a safe-haven premium moves the dollar on the shock, it should move back, at least in part, on its reversal. Two additional military/oil-supply episodes – the twelve-day Israel–Iran war that began on 13 June 2025 and the U.S. strikes of 25 May 2026 – provide further events against which the main results can be checked. Each of these dates is treated as the centre, t_0, of the event window defined in Chapter 5, and announcement dates are used throughout, since these are the points at which the information reaches markets.

4.5 Market Backdrop, 2019–2026

Figures 4.1–4.3 place the episodes in their longer-run context, plotting the broad dollar index, the price of Brent crude, and the Caldara and Iacoviello (2022) geopolitical-risk index over 2019–2026, with the four main events marked on each series. These figures are intended to orient the reader to the level and volatility of each series around the events; the formal measurement of how the dollar responded, and of whether that response differed across episodes, is deferred to Chapter 6.

Figure 4.1: Broad dollar index, 2019–2026, with the four main events marked on the series: the Ukraine invasion (24 February 2022), Liberation Day (2 April 2025), the twelve-day war (13 June 2025), and the Hormuz crisis (28 February 2026). Source: FRED.

Figure 4.2: Brent crude oil price (USD/bbl), 2019–2026, with the same four events marked as in Figure 4.1. Source: FRED.

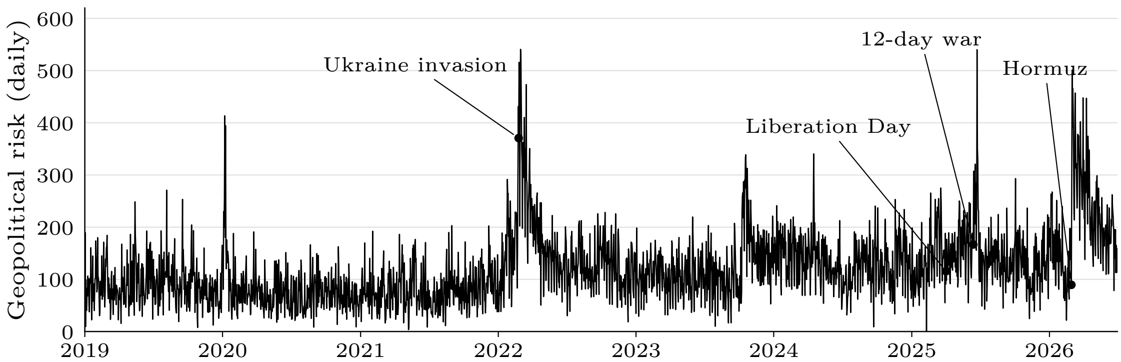

Figure 4.3: Caldara–Iacoviello daily Geopolitical Risk Index, 2019–2026, with the same four events marked as in Figure 4.1. Source: Caldara and Iacoviello's Geopolitical Risk index (Caldara and Iacoviello 2022).

5 Data and Methodology

5.1 Data

5.1.1 Exchange Rates and Financial Variables

The currency panel comprises 32 exchange rates against the U.S. dollar, sampled daily from 1 January 2019 to 30 June 2026. Spot rates are taken from LSEG/Refinitiv at the WM/Reuters (WMR) fixing and exported as a single consistent panel; the Federal Reserve's H.10 release (via FRED) serves as a cross-check. All rates are expressed as foreign currency per U.S. dollar, so that a rise denotes dollar appreciation; the four rates LSEG quotes in the opposite convention (EUR, GBP, AUD, NZD) are inverted accordingly. Daily returns are log changes, $r_{i,t} = \ln(S_{i,t-1}/S_{i,t})$, signed so that a positive value is an appreciation of the foreign currency against the dollar.

The macro-financial block is drawn from FRED: the broad, advanced-economy and emerging-market nominal dollar indices (`DTWEXBGS`, `DTWEXAFEGS`, `DTWEXEMEGS`); the 5- and 10-year Treasury yields (`DGS5`, `DGS10`), the 5-year TIPS yield (`DFII5`) and 5-year breakeven inflation (`T5YIE`); the VIX (`VIXCLS`); Brent and WTI crude (`DCOILBRENTEU`, `DCOILWTICO`); Henry Hub natural gas (`DHHNGSP`); and the S&P 500 (`SP500`). Three series not carried by FRED are taken from LSEG: the gold spot price (`XAU=ZKBZ`, USD/oz), the Euro Stoxx 50 price index (`.STOXX50E`, EUR) and the Dutch TTF front-month natural-gas contract (`TRNLTTFMc1`, EUR/MWh). Indices, equities and commodities enter as log returns; yields and the VIX enter in first differences.

All series are denominated consistently against the U.S. dollar: exchange rates as foreign currency per dollar, and oil and gold in dollars. One consequence requires care. Because commodities are priced in dollars, a dollar movement enters their prices mechanically: a depreciation of the dollar raises the dollar price of oil even if oil is unchanged in every other currency. To separate genuine commodity movements from this denomination effect, Brent and gold are additionally re-expressed in euro—in log returns, $r^{\mathrm{EUR}}_{X,t} = r^{\mathrm{USD}}_{X,t} - r_{\mathrm{EUR},t}$, where $r_{\mathrm{EUR},t}$ is the euro's log appreciation against the dollar—and the euro-denominated series are carried through the event study as a robustness check on denomination.

5.1.2 Geopolitical and Policy Uncertainty Indices

Geopolitical risk is measured by the Caldara and Iacoviello (2022) Geopolitical Risk (GPR) index, used at daily frequency—with its *threat* and *act* sub-indices—and monthly, downloaded from the authors' site. Policy uncertainty is captured by the U.S. daily Economic Policy Uncertainty (EPU) index and the Trade Policy Uncertainty (TPU) index from *policyuncertainty.com*. The dollar and carry risk factors are reconstructed from spot and one-month forward exchange rates following Lustig, Roussanov and Verdelhan (2011): the one-month forward discount $f_{i,t} - s_{i,t}$ proxies the interest-rate differential under covered interest parity; currencies are sorted on this signal each month, and the dollar factor (the average forward-implied excess return) and the carry factor (the high-minus-low portfolio return) are formed. Rebuilding the factors from current market data extends them through the 2026 sample, replacing the course-supplied series that end in 2017.

5.1.3 Portfolio Construction

Two basket schemes are used. The first sorts currencies by their average one-month forward discount over the sample into terciles: the low-yield tercile forms the *safe* (funding) basket and the high-yield tercile the *risky* (carry) basket, with pegged and managed currencies (DKK, HKD, SAR, KWD, CNY) excluded from the sort. This is the classification that underlies the carry factor and the main event-study specification (Figure 5.1). Because the low-yield tercile also contains low-rate Asian currencies that are not conventional safe havens, a second, literature-standard scheme is reported as a robustness check: a *named-haven* safe basket of the yen, Swiss franc and euro against a *risky* basket of the Turkish lira, Brazilian real, South African rand, Mexican peso, Colombian peso and Indonesian rupiah. Independently, currencies are split by net oil-trade position into oil exporters (NOK, CAD, MXN, COP, BRL, SAR, KWD) and oil importers (JPY, KRW, INR, THB, TWD, TRY). All baskets are equally weighted; the equal-weighted average of all currency returns defines the dollar basket (`USD_EW`).

5.2 Event Study Methodology

The main specification is the *constant-mean return* model (MacKinlay 1997; Campbell et al. 1997). For each series i the normal return is its average over an estimation window of $[-140, -21]$ trading days preceding the event, ending where the event

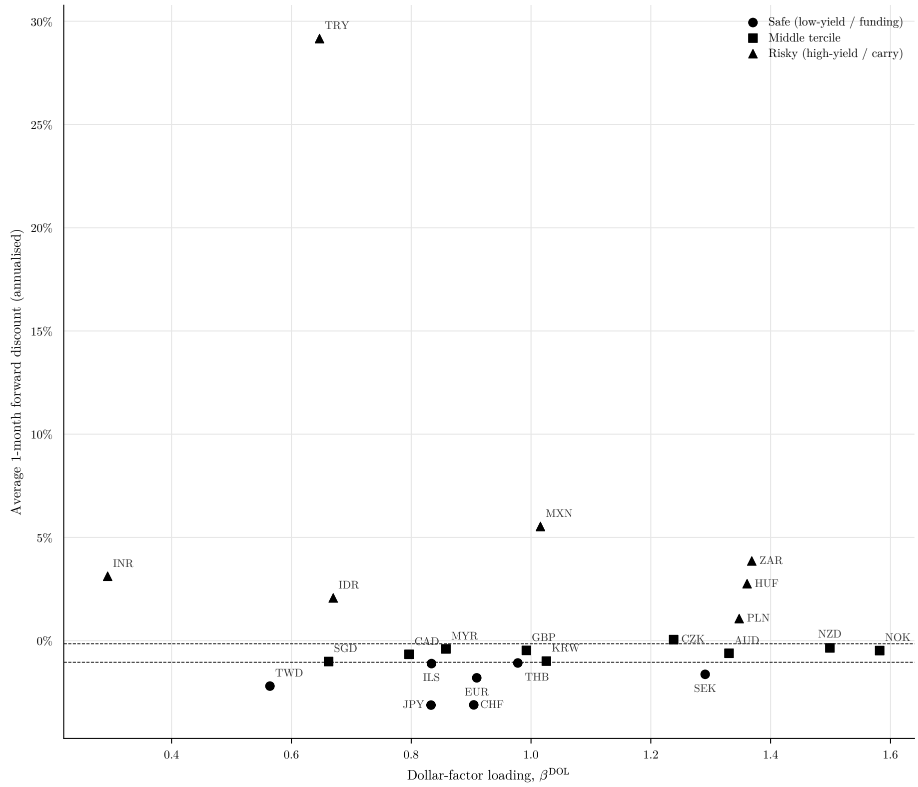

Figure 5.1: Currency classification. Each currency is plotted by its dollar-factor loading β^{DOL} (from a time-series regression of its monthly excess return on the reconstructed dollar and carry factors) against its average one-month forward discount, the signal used for the tercile sort. Dashed lines mark the tercile cut-offs, so marker shape (bucket) and vertical position agree: low forward discount → safe (funding), high → risky (carry).

window begins so that no event-window day enters the estimation of the benchmark; the abnormal return on day t of the event window is the deviation from that mean,

$$\mathrm{AR}_{i,t} = R_{i,t} - \hat{\mu}_i, \qquad \hat{\mu}_i = \frac{1}{120}\sum_{s=-140}^{-21} R_{i,s}, \tag{5.1}$$

and the cumulative abnormal return aggregates these over the window,

$$\mathrm{CAR}_i(t_1, t_2) = \sum_{t=t_1}^{t_2} \mathrm{AR}_{i,t}. \tag{5.2}$$

The event window is $[-20, +20]$ trading days. Because daily financial prices absorb many overlapping influences, the longer the cumulation window the weaker the claim that the measured movement belongs to the event alone (MacKinlay 1997; Kothari and Warner 2007). Inference therefore rests on the short windows: the symmetric $\mathrm{CAR}(-1, +1)$, which also guards against information reaching prices the day before the announcement, and $\mathrm{CAR}(0, +1)$ and $\mathrm{CAR}(0, +5)$ for the announcement reaction. The longer cumulations $\mathrm{CAR}(0, +10)$ and $\mathrm{CAR}(0, +20)$ are reported as robustness checks on the durability of the response rather than as independent identification, and a long horizon $\mathrm{CAR}(0, 50)$—a robustness exhibit on persistence—is estimated on $[-170, -51]$ so the estimation window does not overlap the wider event window. The significance of a single CAR is assessed with $t = \mathrm{CAR}/(\hat{\sigma}_i\sqrt{L})$, where $\hat{\sigma}_i$ is the estimation-window standard deviation of AR_i and $L = t_2 - t_1 + 1$. The difference between two events' CARs is tested with $\Delta = \mathrm{CAR}_A - \mathrm{CAR}_B$ and standard error $\sqrt{L\,(\hat{\sigma}_A^2 + \hat{\sigma}_B^2)}$, the two-sample analogue of the CAR variance in Campbell et al. (1997); independence is warranted because the event windows do not overlap.

The constant-mean model is preferred to the market model because the object of study is the dollar itself. This choice is consequential in currency applications: Kwok and Brooks (1990), comparing competing abnormal-return specifications in foreign-exchange markets, find inference sensitive to both the numeraire and the benchmark model, so event-study practice developed for equities does not carry over automatically. The natural market factor for the currency baskets is the dollar factor, which is endogenous to the response under examination: regressing dollar exchange rates on a dollar factor would absorb the common dollar movement into $\beta_i R_{m,t}$ and leave little in the abnormal return. As a robustness check on *individual* currencies, the market model

$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \varepsilon_{i,t}, \qquad \mathrm{AR}_{i,t} = R_{i,t} - (\hat{\alpha}_i + \hat{\beta}_i R_{m,t}), \tag{5.3}$$

with $R_{m,t}$ the equal-weighted dollar factor, is reported in Appendix 10 and leaves the safe-haven hierarchy unchanged. Where cross-sectional dependence across currencies is a concern, the adjusted test of Kolari and Pynnönen (2010) can replace the plain t-test; because inference here is drawn at the portfolio level around single, non-overlapping event dates rather than by averaging many correlated securities on a common date, the plain t-test is retained.

5.3 Regression Specification: The Safe-Haven Breakdown

The event study measures *whether* the dollar moved; a complementary regression measures *whether its relationship with risk changed.* The classical safe-haven property is that the dollar appreciates when global risk rises, so the daily broad-dollar return is regressed on the daily change in the VIX, interacted with a dummy for each crisis window:

$$r_t^{\mathrm{USD}} = a + b_1\,\Delta\mathrm{VIX}_t + \sum_k \beta_k \left(\Delta\mathrm{VIX}_t \times D_t^k\right) + \sum_k \gamma_k D_t^k + \varepsilon_t, \qquad k \in \{\text{tariff, Hormuz, Ukraine}\}, \tag{5.4}$$

where D_t^k equals one from the event day through twenty trading days after shock k. A positive ΔVIX is a risk-off move and a positive r^{USD} a dollar appreciation, so $b_1 > 0$ is the normal safe-haven reflex; a negative interaction β_k means the reflex weakened during episode k, and the dollar's risk-sensitivity within that window is $b_1 + \beta_k$. Standard errors are Newey–West (HAC, five lags), robust to the volatility clustering of daily returns. Results are reported in Section 6.6.

6 Empirical Results

This chapter presents the thesis's two empirical layers. An event study (Sections 6.1–6.5) measures how the U.S. dollar, the cross-section of currency baskets, and crude oil responded to four shocks; a complementary regression (Section 6.6) then tests whether the dollar's defining relationship with risk held or broke across the episodes. The two layers reinforce one another, reaching the same conclusion by different routes.

The event study isolates the thesis's central contrast: three of the shocks are *U.S.-linked*—the Liberation Day tariff announcement is U.S.-originated, and the twelve-day war and the 2026 Hormuz crisis are military shocks in which the United States is involved as a co-belligerent—while the 2022 Russian invasion of Ukraine is an *external* military/oil shock in which the United States is not the instigator, and which therefore serves as a benchmark for the "classical" safe-haven response. Abnormal returns are computed with a constant-mean model: the normal return of each series is its mean over the estimation window $[-140, -21]$ trading days, the abnormal return is the deviation from that mean, and the cumulative abnormal return $\mathrm{CAR}(t_1, t_2)$ sums abnormal returns over the stated window. Inference rests on the short windows—$\mathrm{CAR}(-1, +1)$, $\mathrm{CAR}(0, +1)$ and $\mathrm{CAR}(0, +5)$—because daily prices absorb many overlapping influences at longer horizons; the ten- and twenty-day cumulations, and the fifty-day persistence exhibit, are read as robustness on the durability of each response (Section 5.2).[1] Significance is assessed with a t-test on the CAR, $t = \mathrm{CAR}/(\hat{\sigma}_{\mathrm{est}}\sqrt{L})$. Tables 6.1 and 6.2 collect the headline numbers; the discussion below reads them event by event.

6.1 CARs: Tariff Shock (Liberation Day)

The April 2, 2025 tariff announcement triggered an immediate depreciation of the dollar. The broad dollar index fell 1.44% on the announcement day and 1.70% over the symmetric $(-1, +1)$ window, both significant at the 1% level (Table 6.1, Panel A). The five-day CAR is temporarily pulled back to -0.10% by the relief bounce that followed the 90-day tariff pause of April 9—the pause itself, dated as its

[1] The constant-mean specification is preferred to the market model here because the object of study is the dollar itself: the natural "market" factor for the currency baskets is the dollar factor, which is endogenous to the response we seek to measure and would absorb the common dollar movement into $\beta\, R_{m,t}$. The market model with the dollar factor is therefore reported only as a robustness check on *individual* currencies (Appendix 10). See Section 5.2.

Table 6.1: The U.S. dollar's response by event and horizon

	Broad USD index — CAR, %					
	Headline windows			Longer horizons (robustness)		
Event	$(-1,+1)$	$(0,+1)$	$(0,+5)$	$(0,+10)$	$(0,+20)$	$(0,+50)$
Panel A. U.S. trade-policy shock						
Liberation Day	-1.70^{***}	-1.44^{***}	-0.10	-3.00^{***}	-3.98^{***}	-6.34^{***}
Tariff pause	-1.48^{**}	-1.48^{***}	-3.07^{***}	-3.44^{***}	-4.51^{***}	-6.74^{***}
Panel B. U.S. military / oil-supply shock						
Twelve-day war	-0.28	$+0.04$	$+0.88$	-0.13	$+0.35$	-0.12
Hormuz crisis	$+1.34^{***}$	$+1.39^{***}$	$+1.53^{***}$	$+2.10^{***}$	$+3.25^{***}$	-0.13
Hormuz ceasefire	-1.33^{***}	-1.14^{***}	-1.72^{***}	-1.65^{**}	-1.47	-0.04
Panel C. External control (non-U.S.)						
Ukraine invasion	$+0.45$	$+0.49$	$+1.03^{*}$	$+1.65^{**}$	$+0.84$	$+3.12^{**}$

Notes. Cumulative abnormal returns of the broad nominal U.S. dollar index (FRED `DTWEXBGS`), in percent, from a constant-mean model. Estimation window $[-140,-21]$ trading days for $h \leq 20$, ending where the event window begins; $[-170,-51]$ for h=50 (pushed back so it does not overlap the wider event window). Inference rests on the headline windows; the longer horizons gauge durability and persistence but absorb confounding events and are read as robustness (Section 5.2). Positive values denote dollar appreciation. Tariff pause and Hormuz ceasefire are sub-events within their parent windows; "Hormuz ceasefire" is the two-week ceasefire of 7 April 2026, distinct from the later 60-day ceasefire of 11 June 2026. The US strikes (25 May 2026) and that 60-day June ceasefire are omitted because the sample ends 30 June 2026, leaving too few post-event days for a stable window. *, **, *** denote significance at the 10%, 5% and 1% level.

Table 6.2: Cross-section of five-day responses: the mirror image

	U.S.-originated			Control
Series, CAR$(0, 5)$ (%)	Liberation Day	Hormuz	Twelve-day war	Ukraine
Broad USD index	-0.10	$+1.53^{***}$	$+0.88$	$+1.03^{*}$
Dollar (equal-weighted)	$+0.21$	$+1.36^{**}$	$+0.83$	$+1.01$
Safe currencies (carry)	$+1.07$	-1.03	-1.04	-1.07^{*}
Risky currencies (carry)	-0.53	-1.97^{***}	-1.03	-2.56^{*}
Oil exporters	-0.80	-0.69	-0.45	$+0.30$
Oil importers	$+0.84$	-1.03^{*}	-1.12	-0.74
Gold	-1.54	-4.54	-1.53	$+1.25$
Euro Stoxx 50	-14.58^{***}	-8.11^{***}	-3.05	-5.96^{**}
Brent crude	-14.32^{***}	$+27.63^{***}$	$+11.25^{***}$	$+13.82^{***}$
WTI crude	-13.14^{***}	$+34.55^{***}$	$+10.14^{**}$	$+14.38^{***}$
Robustness: named safe-haven baskets				
Safe (JPY, CHF, EUR)	$+2.12^{*}$	-1.23	-1.33	-0.69
Risky (TRY, BRL, ZAR, MXN, COP, IDR)	-1.69^{*}	-1.18	-0.36	$+0.24$
Robustness: euro-denominated commodities				
Brent crude in EUR	-15.99^{***}	$+29.21^{***}$	$+12.09^{**}$	$+15.79^{***}$
Gold in EUR	-3.21	-2.97	-0.70	$+3.23^{*}$

Notes. Five-day cumulative abnormal returns (%), constant-mean model, estimation window $[-140, -21]$. Liberation Day is the trade-policy shock; Hormuz and the twelve-day war are the military/oil-supply shocks; Ukraine is the non-U.S. control. For the currency baskets a positive value means the basket *appreciated against the dollar*; for the dollar indices, gold and equities positive means a price increase, and for crude positive means a price increase. *Safe/Risky (carry)* are the bottom and top terciles of the rebuilt forward-discount classification (Section 5.1.3); the first robustness panel reports the literature-standard named-haven baskets. The euro-denominated panel re-expresses the USD-priced commodities in euro (Section 5.1.1) to strip the mechanical dollar-in-the-denominator effect: the crude response survives the change of numéraire, while gold's moves shrink toward zero, consistent with gold trading as a dollar hedge. Gold's CARs are baseline-sensitive given its exceptional 2025–26 trend and are read for sign rather than magnitude (Section 6.4). *, **, *** denote significance at the 10%, 5% and 1% level.

own sub-event, shows the dollar falling a further 1.48% on announcement—before the slide resumes. The longer horizons confirm that the initial reaction understates rather than overstates the move: −3.00% by ten trading days, −3.98% by twenty, and −6.34% by fifty (Section 6.5), indicating that markets read the episode as a durable shift in the U.S. policy regime rather than a transient shock.

The cross-section confirms that the safe-haven premium did not accrue to the dollar but *leaked away from it*. Over the five-day window the named-haven basket—the yen, Swiss franc and euro—appreciated 2.12% against the dollar, the move concentrated in the franc (+3.16%, significant at the 1% level); the broader carry-sorted safe basket, which also contains low-yield Asian currencies, gained a more muted 1.07% (Table 6.2). Either way the flight to quality went to competing havens rather than to the dollar. Crude oil fell sharply—Brent −14.32% and WTI −13.14% over five days, and −15.99% for Brent re-expressed in euro, so the collapse is not an artefact of the weakening dollar in the denominator—European equities sold off hard (Euro Stoxx 50 −14.58%), and the oil-exporter basket slipped 0.80%, consistent with the tariff shock operating through a global *demand*-destruction channel rather than a supply channel. This is the signature of the United States acting as a destabiliser: a U.S.-made shock that simultaneously weakens the dollar, depresses oil and equities, and redirects safe-haven flows to competing currencies.

6.2 CARs: Iran/Hormuz Shock

The two military/oil-supply episodes are reported separately because they differ in both intensity and the role the data assign to the United States.

Twelve-day war (June 2025). The response is muted and statistically insignificant throughout: the broad dollar index moves +0.04% on day one and +0.88% over five days, neither distinguishable from zero (Table 6.1, Panel B). Crude rose meaningfully (Brent +11.25% at five days), but the currency response is essentially flat. This is the "ambiguous actor" case anticipated in the introduction: with the United States a co-belligerent rather than a pure safe harbour, the flight-to-the-dollar reflex and the erosion pressure roughly offset, leaving no significant net move.

Hormuz crisis (February–March 2026). Here the dollar behaves as a classical safe haven. The broad index appreciates 1.39% on impact and 1.53% over five days—both significant at the 1% level—building to +2.10% at ten days and +3.25% at twenty

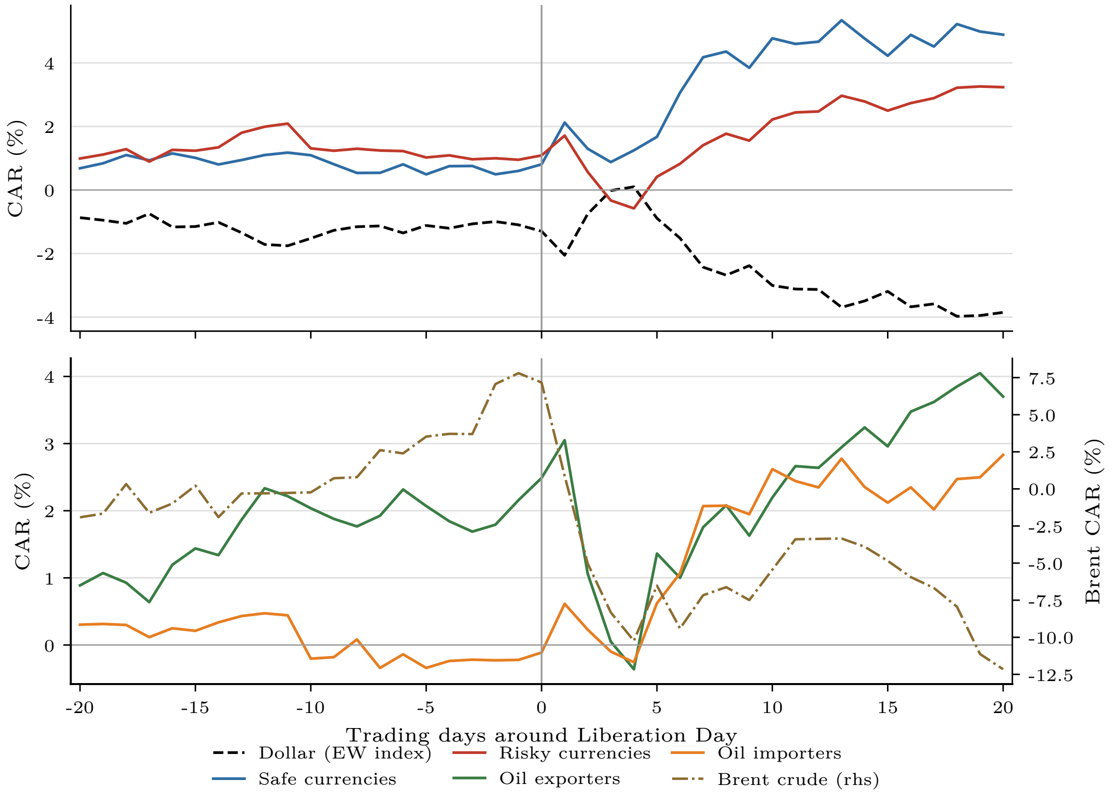

Figure 6.1: Liberation Day (2 April 2025): cumulative abnormal returns over ±20 trading days. Top panel: broad dollar (equal-weighted) and the safe and risky currency baskets. Bottom panel: oil-exporter and oil-importer baskets (left axis) and Brent crude (right axis); WTI is omitted from the figures because Brent prices the bulk of internationally traded crude while the WTI–Brent spread reflects U.S.-local storage logistics (WTI is reported in Appendix 10). The dollar slides while the named-haven currencies bid and crude falls.

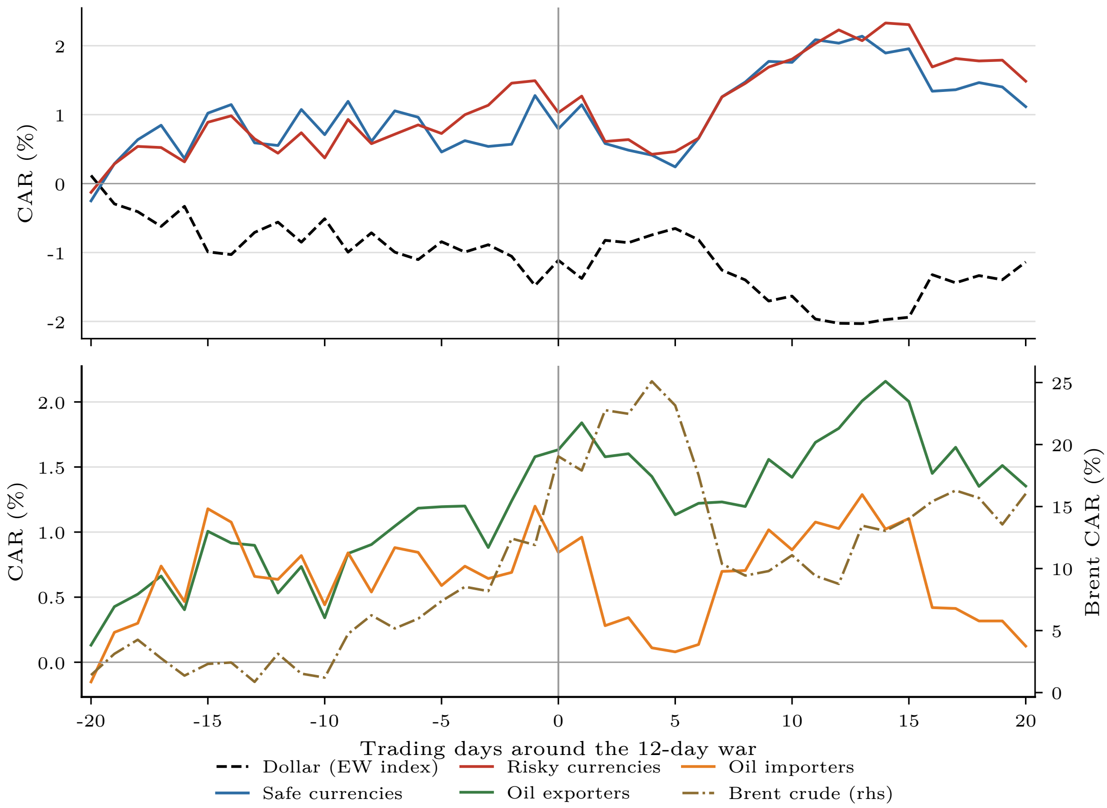

Figure 6.2: Twelve-day war (13 June 2025): cumulative abnormal returns over ±20 trading days. Top panel: dollar and currency baskets; bottom panel: oil-exposure baskets (left axis) and Brent (right axis). Crude rises while the currency response stays flat—the ambiguous-actor case.

(Table 6.1). Crude spikes violently—Brent +27.63% and WTI +34.55% over five days, and +29.21% for Brent in euro, so the spike survives the change of numéraire—and the currency cross-section sorts on oil exposure: the oil-importer basket (yen, won, rupee) depreciates 1.03% against the surging dollar, underperforming the oil-exporter basket, which falls only 0.69% as higher crude cushions exporters. The dollar is bid even though the United States is the originator of the strikes, indicating that for a security/oil-supply shock the safe-haven reflex survives U.S. implication—in sharp contrast to the trade-policy shock.

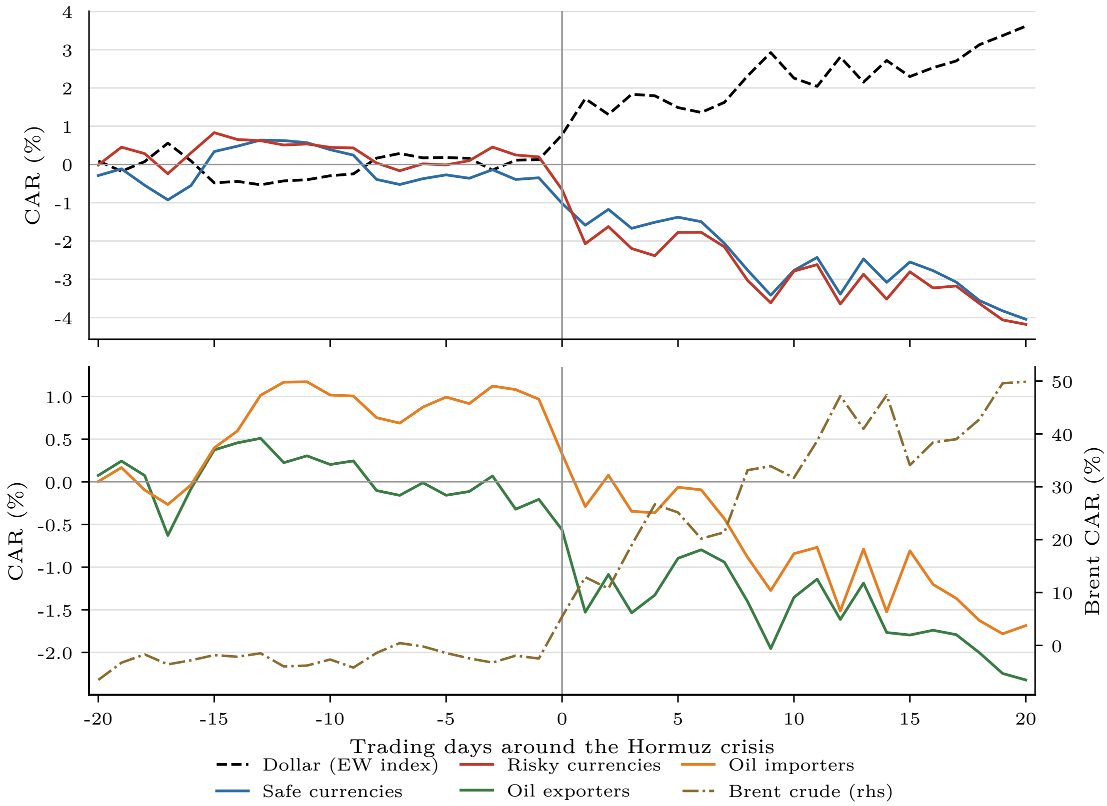

Figure 6.3: Hormuz crisis (28 February 2026): cumulative abnormal returns over ±20 trading days. Top panel: dollar and currency baskets; bottom panel: oil-exposure baskets (left axis) and Brent (right axis). The dollar is bid as crude spikes; oil importers underperform exporters.

Crucially, the Hormuz appreciation is *transitory*. By fifty days the broad-index CAR has returned to -0.13% (insignificant): the entire move unwinds. The clearest evidence that this was a genuine but temporary risk premium comes from the **April 7, 2026 ceasefire**, a within-crisis sub-event marked at +26 days in Figure 6.5. On the two-week ceasefire announcement—which committed to reopening the Strait of Hormuz—the dollar gave back 1.14% on the day and 1.72% over five days, and crude collapsed roughly 22% within ten days. The premium that the threat put *into* the dollar came back *out* on de-escalation, mapping the exchange-rate move directly onto the geopolitical risk it priced. This round-trip is about as close to clean identification as a single-country event study allows.

6.3 Cross-Event Comparison

Reading across Table 6.2, the two U.S.-linked shocks produce a *mirror image*. The trade-policy shock weakens the dollar and depresses oil; the military/oil-supply shock strengthens the dollar and lifts oil. The dollar is therefore not a fixed safe haven or a fixed casualty—its response is conditioned by the *type* of shock, even holding U.S. involvement fixed.

This contrast is not merely descriptive. Table 6.3 tests the difference in the dollar's CAR between events directly, and the separation is immediate: already on the first trading day the broad dollar index is 2.83 percentage points lower under the tariff shock than under the Hormuz crisis ($t = -5.2$) and 1.93 points lower than under Ukraine ($t = -3.5$), both significant at the 1% level. The five-day difference narrows to -1.63 points ($t = -1.7$) as the tariff-pause bounce temporarily retraces the dollar's fall, before the gap re-opens to 5.10 points at ten days ($t = -4.0$) and 7.24 points at twenty ($t = -4.1$); against Ukraine the pattern is the same (-4.65 at ten days, $t = -3.6$). The Hormuz and Ukraine responses, by contrast, share the same sign throughout: at day one the Hormuz bid is somewhat stronger (+0.91 points, $t = 2.0$)—a difference of intensity, not direction—and at five and ten days the two are statistically indistinguishable. The mirror image is thus driven by the trade-policy shock—the only episode in which the dollar moves *against* the classical safe-haven direction, and it does so identifiably from the first day.

The Ukraine benchmark disciplines the interpretation. As an external military/oil shock, it generates the textbook response—the dollar appreciates (+1.65% at ten days, +3.12% at fifty) and crude rises 13.82% (Table 6.2)—and, as just shown, that response shares its sign and, beyond the first day, its magnitude with Hormuz. Two features nonetheless distinguish the external from the U.S.-involved military shock. First, the Ukraine appreciation is durable whereas the Hormuz appreciation reverts, consistent with an external shock conferring a lasting safe-haven bid and a U.S.-involved one only a temporary one. Second, and most importantly for the thesis, the *trade-policy* shock breaks the pattern entirely: the only episode in which the dollar depreciates is the one the United States itself authored on the economic-policy

Table 6.3: Difference in dollar CARs across events (broad USD index, %)

	ΔCAR$(0, h)$, percentage points			
Contrast (A − B)	h=1	h=5	h=10	h=20
Liberation Day − Hormuz crisis	−2.83***	−1.63*	−5.10***	−7.24***
Liberation Day − Ukraine	−1.93***	−1.14	−4.65***	−4.83***
Hormuz crisis − Ukraine	+0.91**	+0.50	+0.45	+2.41*
Liberation Day − Twelve-day war	−1.48**	−0.98	−2.87*	−4.33*

Notes. Difference between two events' cumulative abnormal returns of the broad nominal U.S. dollar index (%), constant-mean model, estimation window $[-140, -21]$. Standard error $\sqrt{L\,(\sigma_A^2 + \sigma_B^2)}$ with $L = h + 1$; events treated as independent (windows do not overlap). A positive value means the dollar rose more (or fell less) under shock A than shock B. */**/***: 10/5/1%.

front. The safe-haven function is intact for geopolitical/oil-supply shocks—even U.S.-involved ones—but it inverts for a U.S. trade-policy shock.

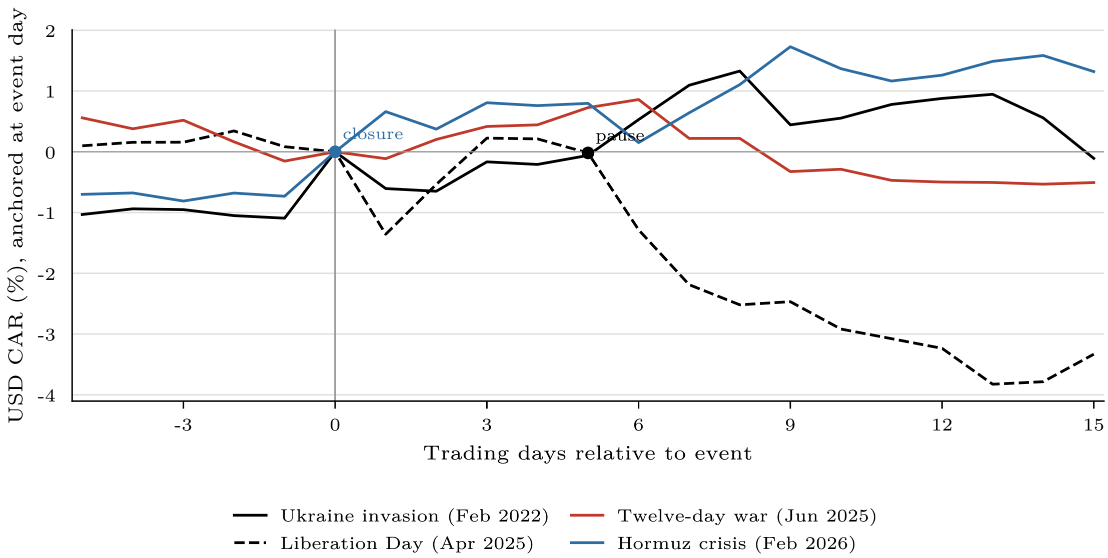

Figure 6.4: Broad dollar CARs across the four headline events, re-anchored on the event day (−5 to +15). The tariff shock is the only episode in which the dollar depreciates; Hormuz and Ukraine trace the classical appreciation.

6.4 Gold and Commodities

The commodity dimension reinforces the demand-versus-supply reading. Brent and WTI move in opposite directions across the two shock types—down ∼13–14% on the tariff shock (a demand shock) and up ∼28–35% on the Hormuz supply shock—and the two benchmarks agree closely, so the result does not depend on the choice of crude reference. Nor does it depend on the currency of denomination: re-expressed in euro, Brent falls 15.99% on the tariff shock and rises 29.21% on Hormuz (Table 6.2), so the oil moves are genuine commodity repricings, not the dollar moving in the denominator (Section 6.5 returns to this). The oil response is the channel through which the cross-section of oil-exporter and oil-importer currencies sorts in Table 6.2.

Gold tells a subtler story than the "universal haven" a first intuition would suggest. Rather than rising around every shock, gold tracks the *inverse of the dollar* across the U.S.-originated episodes: it gains when the tariff shock weakens the dollar (+6.07% at ten days, and +8.85% around the tariff pause) and falls when the Hormuz shock strengthens it (−8.74% at ten days, deepening to −22.3% at twenty). The euro-denominated series sharpens this reading into a test: stripped of the dollar denominator, gold's tariff-window gain essentially vanishes (+0.21% at ten days, insignificant, versus +6.07% in dollars), so most of gold's "rally" around the tariff shock was the dollar falling underneath it. The one departure is the external control: around the Ukraine invasion gold rises *with* the dollar, and does so in euro as well (+6.71% at ten days, significant at the 1% level)—a genuine dual-haven bid rather than a denomination effect. Gold is thus not an unconditional substitute for the dollar but a hedge against it—for the U.S.-made shocks it moves opposite the dollar, reinforcing that the dollar's own conditional response, and not a generic flight to safety, is the organising variable of this chapter; only for the external shock do both havens bid at once. Two caveats temper the magnitudes: gold appreciated roughly 240% over the sample, so the constant-mean baseline is unusually sensitive and the long-horizon gold CARs (e.g. Hormuz at twenty days) overstate the move—the *sign* is the robust feature, not the level—and the single-contributor quote used for the gold fix (LSEG `XAU=ZKBZ`) is noted in Appendix 10.

Equities corroborate the risk-off reading without the dollar's conditionality. The Euro Stoxx 50 falls around every shock—−14.58% on Liberation Day, −8.11% around Hormuz, and −5.96% after the Ukraine invasion (all significant at 5% or better)—so European risk assets sell off whether the shock is a trade war, an oil-supply scare, or an external invasion. The contrast with the dollar is the point: equities are an *unconditional* casualty of geopolitical and policy shocks, whereas the dollar's sign flips with the type of shock. The safe-haven question is therefore specific to the dollar, not a generic property of risk assets.

6.5 Robustness

Four checks support the readings above. First, *longer windows.* Because many things move daily prices at once, the ten- and twenty-day cumulations are not treated as independent identification of the event effect; read as durability checks, they confirm that every headline sign extends rather than reverses (−3.00% and −3.98% for the tariff shock, +2.10% and +3.25% for Hormuz), and the fifty-day horizon (Table 6.1) then separates a level re-rating of the dollar (tariffs, deepening to −6.34%) from a transitory risk premium (Hormuz, fully reverting). The ±50-day figures (e.g. Figure 6.5) plot crude on a secondary axis and mark neighbouring sub-events, making both the persistence and the ceasefire reversal visible. Second, *Brent versus WTI*: the crude responses are quantitatively similar across all events (Table 6.2), so conclusions are invariant to the benchmark. Third, *denomination*: re-expressing the USD-priced commodities in euro (Table 6.2, bottom panel) leaves the crude responses essentially unchanged while shrinking gold's tariff-window move toward zero—the former are genuine commodity repricings, the latter largely the dollar in the denominator (Section 6.4). Fourth, the *market model* with the dollar factor, reported for individual currencies in Appendix 10, leaves the safe-haven hierarchy unchanged.

Two limitations bound the long-horizon results. The ±50-day windows of the two 2025 events overlap (Liberation Day's tail reaches the twelve-day war ∼52 trading days later), so their long-horizon CARs are not fully independent; this is why the persistence figures are read event by event, while the formal cross-event *difference* tests (Table 6.3) use horizons from one to twenty days, at which the windows of the compared pairs do not overlap. And the two most recent shocks—the May 2026 U.S. strikes and the June 2026 ceasefire—are omitted from the longer horizons because the sample ends 30 June 2026, leaving too few post-event trading days for a stable window.

6.6 Regression Evidence on the Safe-Haven Breakdown

The event study above shows *that* the dollar moved in opposite directions across the two shock types. This section asks the sharper question directly: did the dollar's defining safe-haven property—appreciating when global risk rises—actually hold or break in each episode? The dummy-interaction regression specified in Section 5.3

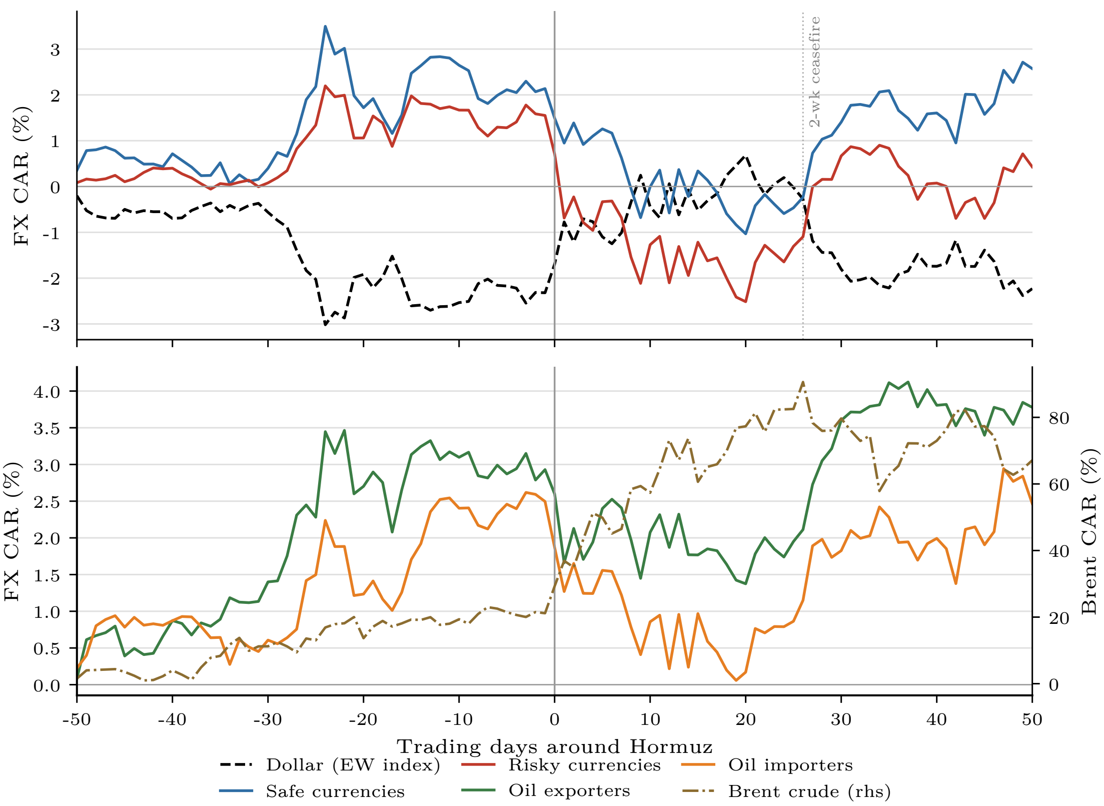

Figure 6.5: Hormuz crisis, ±50 trading days. Top panel: dollar and currency baskets; bottom panel: oil-exposure baskets (left axis) and Brent (right axis). The dollar's appreciation reverts by +50; the April 7 two-week ceasefire (marked) both crashes crude and unwinds the dollar premium.

answers it with ordinary least squares and heteroskedasticity-robust standard errors, requiring none of the modelling assumptions of a conditional-correlation (GARCH) approach.

Table 6.4: Did the dollar's safe-haven reflex survive each shock? (OLS, HAC standard errors)

	(A) VIX	(B) VIX + GPR	(C) VXY	(D) GPR + TPU
ΔVIX (normal)	+0.069*** (+7.0)	+0.070*** (+7.1)	—	—
× Tariff	−0.071*** (-2.6)	−0.067*** (-2.7)	—	—
× Hormuz	+0.057 (+1.3)	+0.062 (+1.3)	—	—
× Ukraine	+0.165*** (+4.8)	+0.168*** (+4.0)	—	—
ΔGPR (normal)	—	+0.013** (+2.0)	—	+0.011* (+1.7)
× Tariff	—	+0.154* (+1.8)	—	+0.180* (+1.8)
× Hormuz	—	+0.002 (+0.0)	—	−0.011 (-0.3)
× Ukraine	—	−0.000 (-0.0)	—	−0.025 (-0.5)
ΔVXY (normal)	—	—	+0.107*** (+6.9)	—
× Tariff	—	—	−0.127*** (-3.2)	—
× Hormuz	—	—	+0.161*** (+3.9)	—
× Ukraine	—	—	+0.117*** (+4.3)	—
ΔTPU (normal)	—	—	—	−0.003 (-0.5)
× Tariff	—	—	—	−0.022 (-1.1)
× Hormuz	—	—	—	+0.051 (+0.5)
× Ukraine	—	—	—	+0.032 (+0.1)
Window dummies	Yes	Yes	Yes	Yes
N	1954	1954	1954	1954
R^2	0.057	0.061	0.137	0.009

Notes. OLS of the daily broad-dollar return (percent) on daily risk shocks, their interactions with crisis-window dummies, and the window (level) dummies; Newey–West (HAC, 5 lags) standard errors, t-statistics in parentheses. Each risk shock is standardized to unit variance, so coefficients read as the percent dollar move on a one-standard-deviation risk-off day. Crisis windows run from the event day to +20 trading days. VIX: CBOE (FRED `VIXCLS`); GPR: Caldara and Iacoviello (2022) daily index; VXY: J.P. Morgan VXY Global FX implied volatility (LSEG `.JPMVXYGL`, mid); TPU: Caldara et al. (2020) daily index, current vintage (downloaded 23 July 2026). A positive slope is safe-haven behaviour (risk up → dollar up); a negative interaction weakens it. */**/***: 10/5/1%.

The result is unambiguous (Table 6.4). On normal days the dollar is a safe haven: the coefficient on the change in the VIX is positive and strongly significant ($b_1 = 0.0003$, $t = 7.0$), so a risk-off jump in volatility lifts the dollar. During the **tariff** window this reflex *disappears*: the interaction is equal and opposite ($\beta_{\text{tariff}} = -0.0003$, significant at the 1% level), so the within-window sensitivity $b_1 + \beta_{\text{tariff}}$ is essentially zero. The dollar stopped behaving as a haven precisely when the United States was the source of the shock. During the **Hormuz** window the interaction is small and statistically insignificant ($t = 1.3$), leaving the safe-haven relationship unchanged from normal; during the **Ukraine** window it is, if anything, *stronger* ($\beta = +0.0008$, $t = 4.8$). The window (level) dummies echo the event study—the average dollar drift is negative in the tariff window and positive in the Hormuz and Ukraine windows.

This regression reaches the event study's conclusion by a different and fully transparent route: the dollar's safe-haven function broke down for the U.S. trade-policy shock but remained intact for the geopolitical and oil-supply shocks. Because it rests on a single, interpretable least-squares specification with robust standard errors, it is the test most directly tied to the central hypothesis—and it delivers that test at a fraction of the modelling and estimation risk of a time-varying-volatility alternative.

7 Discussion

This chapter draws the event-study and regression results together, relates them to the literature, and sets out their implications and limits.

7.1 Findings in Light of the Literature

The results speak directly to the five literatures set out in Chapter 2. The sharpest verdict is for the *dollar-erosion* strand. Kamin (2025) and Jiang et al. (2026) read the 2025 tariff episode as evidence that the dollar has stopped behaving as a safe haven, and Hassan et al. (2025) supply the theory: a trade war that isolates U.S. goods markets erodes the currency's safety premium. The findings here both confirm and bound that claim. They confirm it for the tariff shock—the dollar depreciates and its risk sensitivity collapses to zero—but they bound it by showing that the erosion is not a general weakening: the very same dollar appreciates around the contemporaneous Hormuz shock. What the dollar-erosion papers identify as a regime change is, on this evidence, specific to U.S.-originated economic-policy shocks rather than a property of the post-2025 dollar as such.

For the *safe-haven* tradition (Ranaldo and Söderlind 2010; Habib and Stracca 2012; Bock and de Carvalho Filho 2015) the evidence is corroborative. The dollar performs as that literature predicts around both military/oil-supply shocks, appreciating as risk rises, and the formal difference test cannot separate the U.S.-implicated Hormuz response from the external Ukraine benchmark—so the haven bid survives even U.S. involvement when the shock is a security one. The conditional, crisis-contingent definition of Baur and Lucey (2010) is thus vindicated for the dollar: its haven status is real but state-dependent. That same framing organises the *gold* result. Gold hedges against the dollar rather than serving as a universal safe haven—it bids when the tariff shock weakens the dollar and falls when the Hormuz shock strengthens it, rising alongside the dollar only for the external Ukraine shock—exactly the state-dependence Baur and Lucey's distinction implies.

The *commodity-currency* literature (Y.-C. Chen and Rogoff 2003; Kilian 2009) is borne out selectively: oil-exporter currencies outperform importers only around the supply-driven Hormuz shock, the asymmetry Kilian's typology implies, while the demand-driven tariff decline in oil produces no comparable sorting. Finally, the *geopolitical-risk* literature (Caldara and Iacoviello 2022) supplies both the measurement and a mechanism—the within-crisis ceasefire reversal, which unwinds the dollar premium as the threat recedes, is the threat-versus-realisation distinction Caldara and Iacoviello emphasise, mapped onto the exchange rate.

7.2 The Dollar's Safe Haven Is Conditional on Shock Type

The central finding is that the dollar's safe-haven response is conditioned by the *type* of shock, not merely by who causes it. The dollar appreciated around both military/oil-supply shocks—the external Ukraine invasion and the U.S.-involved Hormuz crisis—and the formal difference test cannot distinguish the two responses, even though the United States authored one and merely witnessed the other. Yet it depreciated around the one purely economic-policy shock it created, the Liberation Day tariffs. Origin alone does not explain the pattern; the nature of the shock does. This refines the "dollar-erosion" reading (Ostry et al. 2025; Kamin 2025): the safe-haven function is not weakening in general but is specifically suspended when the United States is itself the source of an economic-policy shock, while it survives intact—reversion notwithstanding—for the security shocks that the safe-haven tradition has always emphasised (Ranaldo and Söderlind 2010; Habib and Stracca 2012).

The regression sharpens this further. Estimated over the full sample, the dollar's sensitivity to risk is significantly positive on normal days—the textbook safe-haven reflex—but the interaction term shows it collapsing to essentially zero inside the tariff window, while it remains intact during the Hormuz and Ukraine windows. The breakdown is thus not merely a difference in the *level* of the dollar across episodes but a change in its very *relationship with risk*: during the U.S. trade-policy shock the dollar stopped trading like a safe haven, and only then.

Read against the hypotheses of Section 3.4, the evidence adjudicates all three. **H1 is supported**: the dollar appreciated around both military/oil-supply shocks, depreciated around the trade-policy shock, and the difference-in-CARs test separates the tariff episode from both others while leaving Hormuz and Ukraine indistinguishable. **H2 is supported**: gold rose in both kinds of episode (Section 7.3), so the conditionality is a property of the dollar, not of safe assets generally. **H3 is supported**: the exporter–importer spread carries the sign of the crude response in each episode, positive under Hormuz and Ukraine and negative under the tariff shock (Section 7.3).

7.3 Gold and the Oil Channel

Gold does not behave as the universal safe haven that a first reading of Baur and Lucey (2010) would predict. Instead it tracks the inverse of the dollar across the U.S.-originated shocks—rising when the tariff shock weakened the dollar, falling when the Hormuz shock strengthened it—and bids alongside the dollar only for the external Ukraine shock. Gold is thus better described as a hedge *against* the dollar than as a substitute for it, which reinforces, rather than competes with, the chapter's main message: the dollar's own conditional move is the organising variable, and gold prices it from the other side.

The oil channel operates as the commodity-currency literature predicts, but only for the supply shock. During the Hormuz crisis, oil-importer currencies underperformed oil-exporters as crude surged, consistent with Y.-C. Chen and Rogoff (2003) and Kilian (2009); during the demand-driven tariff decline the sorting was muted. The yen illustrates the resulting tension: a classical safe haven that is also a large oil importer, it is pulled in opposite directions during an oil-supply shock, which is part of why the basket responses around Hormuz are smaller and noisier than the dollar-index response.

7.4 Policy Implications

Three implications follow. For reserve managers and central banks, the dollar's safe-haven status cannot be treated as unconditional: hedging programmes calibrated on the assumption that the dollar always rallies in risk-off episodes will misfire precisely when the shock originates in U.S. economic policy. For portfolio managers, diversification should be thought of across *shock types*, not only across assets—gold and competing-haven currencies, not the dollar, carried the flight to quality in the tariff episode. For the broader debate on the dollar's reserve role (Obstfeld and Zhou 2023; Jiang et al. 2026), the evidence is a caution against over-reading any single episode: the 2025 tariff shock did dent the dollar, but the 2026 security shock showed the haven bid still firmly in place.

7.5 Limitations

Several caveats bound the results. The two most recent events—the May 2026 U.S. strikes and the June 2026 ceasefire—have too few post-event trading days for the longer windows, so they are reported only at short horizons. The regression's crisis windows are defined around the events rather than by a structural break test, and the VIX–dollar link it measures is a reduced-form association, not a causal channel. Gold's exceptional 2025–26 trend makes its constant-mean CARs sensitive in level, so they are read for sign rather than magnitude. As a single-country event study around a small number of recent, partly overlapping episodes, the design trades breadth for the clean U.S.-versus-external contrast; and without forward/basis data beyond the one-month tenor, covered-interest-parity deviations are left for future work.

8 Conclusion

This thesis asked whether the U.S. dollar's safe-haven response depends on the type of shock that triggers a crisis, even when the United States is implicated in the shock itself. Two recent episodes provided an unusually clean test: the 2025 Liberation Day tariffs, a purely U.S. economic-policy shock, and the 2026 Iran/Hormuz crisis, a geopolitical oil-supply shock, with the 2022 Russian invasion of Ukraine as an external control.

The evidence gives a clear answer. The two shock types produced a mirror image. The dollar depreciated around the tariff shock—over three and a half percent on the broad index within twenty trading days—while it appreciated around the Hormuz crisis and around Ukraine, and a formal difference test confirms the tariff response is significantly distinct from both security episodes, which are themselves statistically indistinguishable from one another. The dollar's haven status is therefore conditional on the *nature* of the shock, not on its origin: it survives a U.S.-involved security shock but inverts for a U.S.-involved economic-policy shock. A complementary regression confirms the breakdown statistically: the dollar's safe-haven sensitivity to risk—significantly positive on normal days—collapses to zero in the tariff window, while holding through the Hormuz and Ukraine episodes. Two further results round out the picture: gold acts as a hedge against the dollar rather than as a universal safe haven, and the oil channel sorts currencies only for the supply-driven shock.

The thesis contributes the first systematic side-by-side comparison of the Liberation Day and Iran/Hormuz episodes for currency markets, carries a gold and oil benchmark through the same design to connect the safe-haven and commodity-currency literatures, and extends standard cross-sectional currency tools to mid-2026 on a consistent WMR dataset.

Several extensions suggest themselves. As the geopolitical situation continues to evolve, the sample can be extended so that the most recent episodes acquire full post-event windows. Richer forward and basis data would allow covered-interest-parity deviations to be brought into the analysis, and intraday data would tighten the identification around the announcement windows. A dynamic conditional-correlation (DCC-GARCH) model of the dollar–VIX relationship would test the safe-haven-breakdown hypothesis directly and is the natural next layer. Finally, a formal model of shock-dependent safe-haven demand—in which the currency of the shock's originator loses its haven premium for policy shocks but not for security shocks—would give the empirical pattern documented here a theoretical foundation.

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Appendix

1 Currency Universe and Classification

Table 1 lists the 32 currencies in the panel with their carry-sorted bucket (low/middle/high forward-discount tercile, pegged and managed currencies excluded from the sort) and their oil-trade role. The named safe-haven robustness baskets used in Chapter 6 are the yen, Swiss franc and euro (safe) against the Turkish lira, Brazilian real, South African rand, Mexican peso, Colombian peso and Indonesian rupiah (risky).

2 Robustness

Market model. Alongside the constant-mean specification, the event study computes market-model abnormal returns, using the equal-weighted dollar factor as the market return (Code/`03_event_study.py`, method `market_model`). Table 2 reports the basket-level results: the safe-haven ordering is unchanged relative to the constant-mean results in Chapter 6.

Benchmark and persistence figures. Figure 1 shows the full portfolio response around the external Ukraine benchmark at the identification window; in the main text its dollar path appears only inside the cross-event overlay (Figure 6.4). Figure 2 extends the Liberation Day window to ±50 days: the tariff drift deepens rather than reverts, the mirror image of the transitory Hormuz premium in Figure 6.5.

Regression robustness. The safe-haven regression of Section 6.6 uses Newey–West (HAC) standard errors, so its inference is robust to the autocorrelation and volatility clustering of daily returns—the main reason a simple least-squares interaction is preferred here to a conditional-correlation model. The crisis windows are the same event-day-to-+20 windows used in the event study, so the regression and the event study rest on a common, transparent definition of each episode.

Table 1: Currency classification

Code	Currency	Carry bucket	Oil role
AUD	Australian dollar	middle	—
BRL	Brazilian real	high (risky)	exporter
CAD	Canadian dollar	middle	exporter
CHF	Swiss franc	low (safe)	—
CLP	Chilean peso	no 1M forward	—
CNY	Chinese yuan	pegged/managed	—
COP	Colombian peso	no 1M forward	exporter
CZK	Czech koruna	middle	—
DKK	Danish krone	pegged (EUR)	—
EUR	Euro	low (safe)	—
GBP	Pound sterling	middle	—
HKD	Hong Kong dollar	pegged (USD)	—
HUF	Hungarian forint	high (risky)	—
IDR	Indonesian rupiah	high (risky)	—
ILS	Israeli shekel	low (safe)	—
INR	Indian rupee	high (risky)	importer
JPY	Japanese yen	low (safe)	importer
KRW	South Korean won	middle	importer
KWD	Kuwaiti dinar	pegged	exporter
MXN	Mexican peso	high (risky)	exporter
MYR	Malaysian ringgit	middle	—
NOK	Norwegian krone	middle	exporter
NZD	New Zealand dollar	middle	—
PEN	Peruvian sol	no 1M forward	—
PLN	Polish zloty	high (risky)	—
SAR	Saudi riyal	pegged	exporter
SEK	Swedish krona	low (safe)	—
SGD	Singapore dollar	low (safe)	—
THB	Thai baht	low (safe)	importer
TRY	Turkish lira	high (risky)	importer
TWD	Taiwan dollar	low (safe)	importer
ZAR	South African rand	high (risky)	—

Notes. Carry buckets are terciles of the average one-month forward discount (Section 5.1.3); pegged/managed currencies and the three currencies without a clean one-month forward (CLP, COP, PEN) are excluded from the carry sort. Oil role follows the net oil-trade classification.

Table 2: Market-model CARs of the currency baskets (percent). Abnormal returns are residuals from a regression of each basket on the equal-weighted dollar factor estimated over the same $[-140, -21]$ window (Section 5.2); the safe-haven hierarchy of the constant-mean results is unchanged. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	Ukraine		Liberation Day		12-day war		Hormuz	
	(0,5)	(0,20)	(0,5)	(0,20)	(0,5)	(0,20)	(0,5)	(0,20)
Safe currencies	-0.31	-1.73**	1.31***	1.17*	-0.13	0.21	0.67**	0.67
Risky currencies	-0.89	-0.21	-0.33	-0.48	-0.26	0.31	-0.73***	-1.20***
Oil exporters	1.26***	2.29***	-0.62	-0.77	0.34	0.10	0.61	1.23*
Oil importers	0.51	-1.97	1.00***	0.95	-0.49	-0.82	-0.00	-0.02
Carry (risky−safe)	-0.57	1.52	-1.63***	-1.65	-0.13	0.10	-1.40***	-1.87**

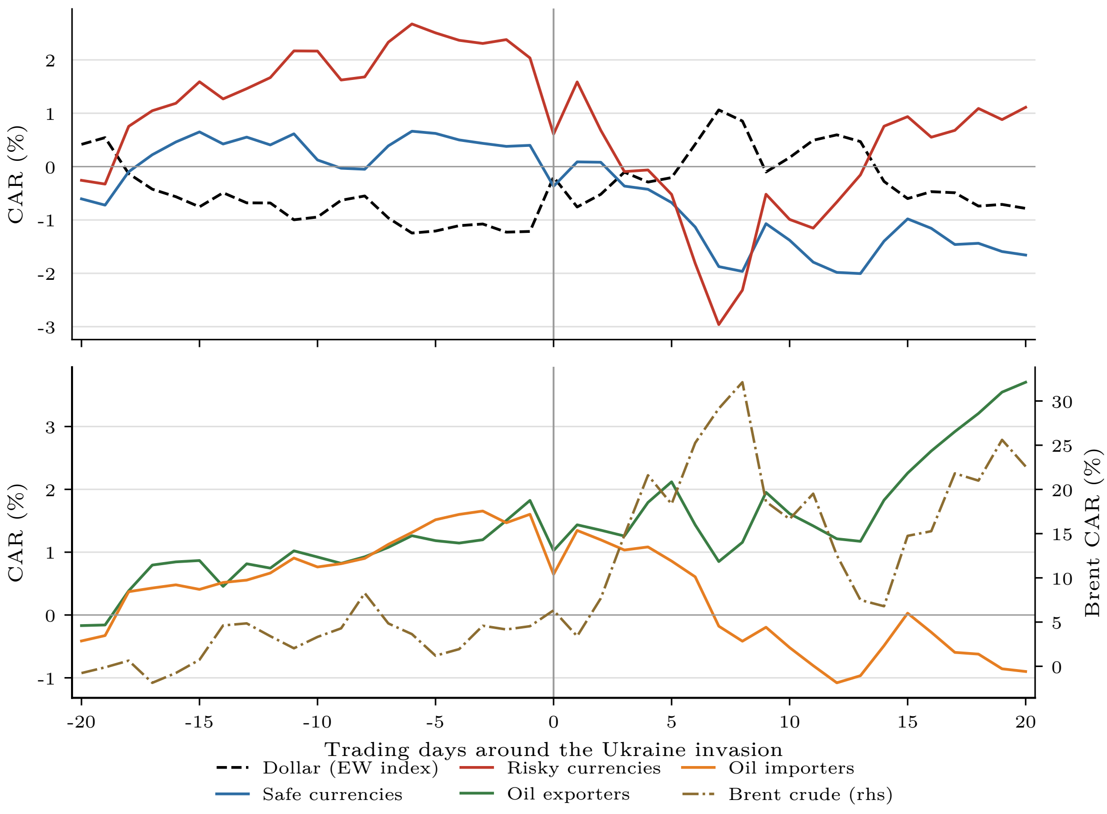

Figure 1: Ukraine invasion (24 February 2022): cumulative abnormal returns over ± 20 trading days. Top panel: dollar and currency baskets; bottom panel: oil-exposure baskets (left axis) and Brent (right axis).

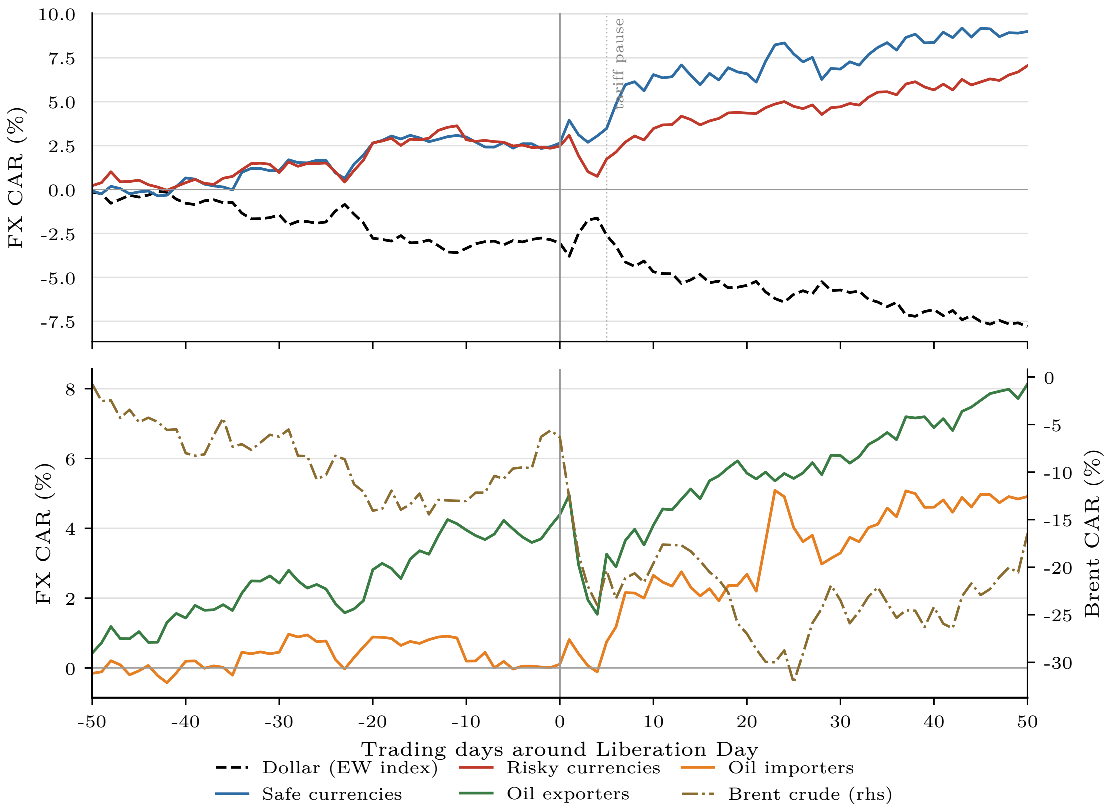

Figure 2: Liberation Day, ±50 trading days. Top panel: dollar and currency baskets; bottom panel: oil-exposure baskets (left axis) and Brent (right axis). The post-announcement dollar decline deepens over the horizon; the April 9 tariff pause (marked) interrupts but does not reverse it.

Affidavit

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